CTIS411 Senior Project

2025-2026

Initial Plan

An Eye

Team 12

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Change History

File Name	Document Type	Deliverable	Version	Submission Date
Team12_InitialPlan_V1	MS Word 365	1	1	06.10.2025
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Change List

- The Executive Summary was extended.
- 5 New Functional Requirements were added.
- FR3-2 was moved to Non-Functional requirements as NFR-4.
- The Context Diagram was re-drawn.
- The Software Development Process Model part rewritten according to Incremental Model with Agile Principles.
- SDLC visualization was drawn.
- Two Risks that were not related to the project were removed. Two project risks were added.
- Project Risks were quantitatively assessed.

Project Details

Project Name	**An Eye**
Software Product Name	**ObservAI**
Company Name	**No sponsor company**
Academic Advisor	**Leyla Sezer**
Github URL	**https://github.com/observaianeye/ObservAI**
WEB page	
Software Product Information	ObservAI is an AI-powered analytics system designed to help café and restaurant managers better understand their customers and optimize their spaces. Using camera-based analysis, the system observes how people move within the area, identifies crowded zones, estimates basic demographics such as age and gender ranges, and visualizes customer activity through live heatmaps. These insights are presented on an intuitive web dashboard where managers can monitor customer flow, compare different time periods, and make data-informed decisions about layout or service improvements. The system's purpose is to transform everyday visual data into clear and actionable information that supports smarter management and smoother operations. ObservAI also prioritizes user privacy and ethical use of AI by processing all data anonymously and without storing personal or identifying visuals.

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Individual Contributions Overview

Name, Surname	Summary of Contributions to the Initial Plan Document
Faik Emre Partal	Contributed to the initial planning of the document by defining general ideas, key requirements, and the overall project scope.
Halil İbrahim Atalay	Wrote content related to the software development process model and the product purpose and helped with overall document content
Mevlüt Yılmaz	Refactored and edited the documentation, correcting formatting issues, abbreviations, and general language errors.
Semih Şengül	Authored sections on project communication and project assumptions.
Uğur Ayber Karakış	Designed and added diagrams to enhance the visual presentation and performed minor content refactoring.

Executive Summary

The ObservAI Initial Plan outlines the foundation for developing an intelligent, data-driven management system tailored for cafés and restaurants. The project's vision is to transform everyday visual activity into meaningful insights that help managers make informed, timely, and efficient decisions without requiring technical expertise. ObservAI employs AI-powered computer vision to interpret real-time camera data, detect visitor patterns, estimate demographic ranges, and visualize customer flow through interactive heatmaps. These analytics are presented on a user-friendly web dashboard that supports decision-making in space utilization, service optimization, and crowd management. The Initial Plan defines the objectives, scope, deliverables, risks, and development roadmap that guide the project throughout both academic semesters. The chosen software process is the Incremental Development Model enhanced with Agile principles, ensuring that the product evolves through iterative cycles of implementation, feedback, and improvement. Each increment introduces measurable functionality—from authentication and dashboard prototypes to AI integration and insight generation—allowing the team to maintain progress and adaptability under realistic academic timelines. ObservAI's design prioritizes privacy and ethical use of artificial intelligence. The system processes all camera data anonymously, strictly following KVKK and GDPR standards to prevent any personal identification or misuse. The project's functional requirements emphasize usability, accuracy, and reliability, while non-functional ones ensure performance, scalability, and data security. Risk management practices, including regular reviews, dependency tracking, and mitigation planning, are integrated into each stage to safeguard delivery quality. Ultimately, this Initial Plan serves as the strategic backbone of ObservAI's development. It transforms an academic concept into a structured, feasible roadmap aimed at delivering an accessible, ethical, and innovative analytics platform by June 2026. Through a combination of artificial intelligence, web technologies, and teamwork, ObservAI aspires to demonstrate how modern cafés and restaurants can leverage technology to make smarter, data-driven decisions.[1]

[1] ChatGPT 5 Prompt: **"Can you give exactly one page executive summary for this Document? "**

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Abbreviations

AI	Artificial Intelligence
CTIS	Information Systems and Technologies
FReq	Functional Requirement
GDPR	General Data Protection Regulation
HTTPS	Hypertext Transfer Protocol Secure
IP	Initial Plan
KVKK	Kişisel Verilerin Korunması Kanunu (Personal Data Protection Law)
NFReq	Non-Functional Requirement
PMBOK	Project Management Body of Knowledge
POS	Point of Sale
SDD	Software Design Description
SDLC	Software Development Lifecycle
SPMP	Software Project Management Plan
SRS	Software Requirements Specification
TLS	Transport Layer Security
UI	User Interface
WBS	Work Breakdown Structure

Initial Plan

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1. Product Purpose

ObservAI is designed as a real-time analytics and visualization platform that transforms camera data into actionable insights for cafés and restaurants. By using AI-powered computer vision, the system enables managers to monitor customer activity, estimate demographic patterns, and analyze in-store dynamics. Cameras detect how many people enter or pass by, estimate their gender and age ranges, identify busy areas, and measure table occupancy durations.

Through its centralized web dashboard, ObservAI presents these insights clearly and interactively, helping managers understand customer flow and optimize space usage without manual observation. The platform focuses solely on AI-driven visual analytics rather than broader operational management tasks. Its purpose is to enhance business awareness through data-based decision-making.

ObservAI aims to improve efficiency, customer experience, and profitability by highlighting behavioral patterns and peak activity periods. With real-time analysis, heatmaps, and demographic insights combined in a single system, it helps cafés minimize wait times, balance seating, and maintain consistent service quality.

2. Product Requirements

The users of ObservAI shall primarily be café and restaurant managers or owners. The system shall have a single managerial role with secured access and an intuitive interface. Each user shall log in using a verified email and password. To ensure account authenticity, the sign-up process shall include email verification, and the dashboard shall appear after successful login.

The core component of the system shall be the operations dashboard (FReq1), which displays AI-driven insights about customer activity and in-store behavior. By using computer vision, ObservAI detects and counts visitors, estimates demographics such as age ranges and gender distribution, identifies queues, and generates real-time heatmaps that visualize customer density and movement patterns. These analytics are presented in a clean, interactive view where the manager may switch between different visual modes such as bar charts, line graphs, and spatial heatmaps without leaving the dashboard (FReq2).

The notification system (FReq3) shall alert managers about specific operational situations such as long queues, unexpected crowd build-ups, or unusual traffic patterns. This feature helps management take timely actions to optimize service and reduce waiting times.

The AI engine (FReq4) shall analyze behavioral trends and occupancy data to produce simple operational insights. For instance, it may suggest rearranging seating layouts or adjusting service timing during peak hours. The insights aim to inform decisions with data-driven observations rather than directive recommendations.

ObservAI's non-functional requirements ensure that the platform is fast, reliable, and secure. The system shall be accessible via web browsers on both desktop and mobile devices (NFReq1), allowing managers to monitor operations remotely. Daily database backups (NFReq2) shall ensure recovery in case of system failures. All communications between the client and server shall be encrypted using HTTPS/TLS (NFReq3). Customer data obtained via cameras shall remain anonymized and processed in full compliance with KVKK and GDPR policies (NFReq4).[2]

ObservAI focuses purely on real-time AI analytics and visualization for managerial awareness. It also excludes features such as facial recognition, voice analysis, or online ordering. In

[2] ChatGPT 5 Prompt: **"Can you add the Non-Functional Requirement part that we wrote in the SRS document . Write it clearly and shortly. "**

conclusion, ObservAI is an advanced analytics system designed to turn live camera data into practical insights for café and restaurant managers, while ensuring data privacy.

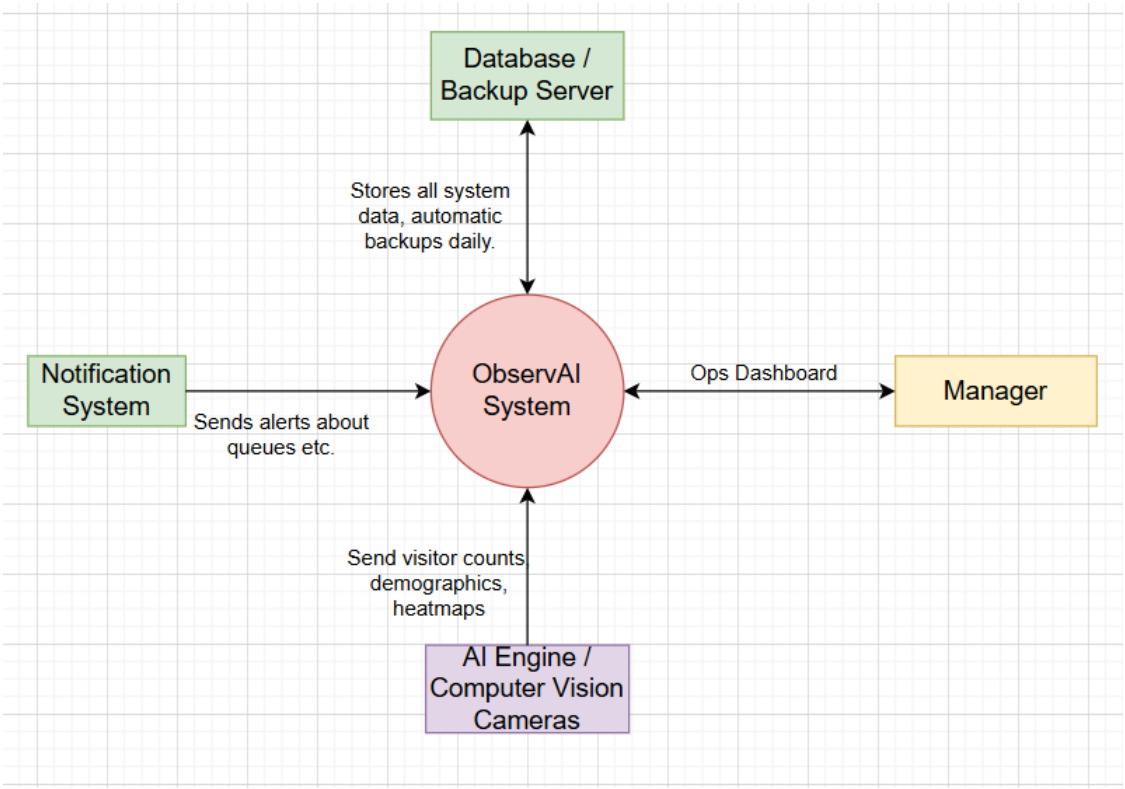

Figure 1 Context Diagram

3. Software Development Process Model

We will use the Incremental Software Development Model in our project, while adopting several Agile principles to maintain flexibility and continuous improvement. The Incremental Model allows the system to be developed and delivered in small, functional parts called increments. Each increment adds new capabilities to the system, and together they gradually build the final product. By applying Agile principles such as iterative development, continuous feedback, and collaboration, each increment will go through short development cycles involving planning, implementation, testing, and evaluation.[3] This approach enables us to adapt to changes and receive ongoing input from our advisor and stakeholders, ensuring that the system evolves in line with real needs. We chose the Incremental model supported by Agile principles because our project requirements may change, and we want a flexible yet structured process to handle those changes effectively.

In the first semester, we agreed that each increment will take 2 weeks and at the end of each increment we will meet with our advisor to review what we did in the past 2 weeks. At those meetings we will give feedback to our teammates and also our advisor will give us feedback if there is any part that has to change. At the end of those meetings, we assign new tasks to each of our group members.

In the second semester, since we will start writing code we have planned to change the gap between our meetings by 1 week only. These weekly meetings will also serve as planning and retrospective sessions, aligning the team's work and maintaining consistent progress across increments. These meetings will also function as sprint planning and retrospective sessions, ensuring continuous alignment within the team.

Regular feedback loops will be maintained with the supervisor throughout both semesters, focusing on both technical and design aspects of the system. Each increment will undergo

[3] ChatGPT 5 Prompt: **"Change this part such the way that we use incremental model by adopting agile principles."**

internal testing for usability, performance, and functionality before integration with the main system.[4]

Using the Incremental Model with Agile principles allows us to respond rapidly to feedback while building the system step by step, ensuring steady progress and adaptability. However, using a model that is so open to feedback seems to pose a risk that could make it difficult for our project to be established on an unchangeable basis.

All team members already have experience with Agile-based development processes such as Scrum and Kanban, which will make it easier to apply Agile practices within the Incremental Model framework. Overall, Incremental with Agile principles enables iterative development, continuous feedback, and rapid adaptation, leveraging our team's prior experience.[5]

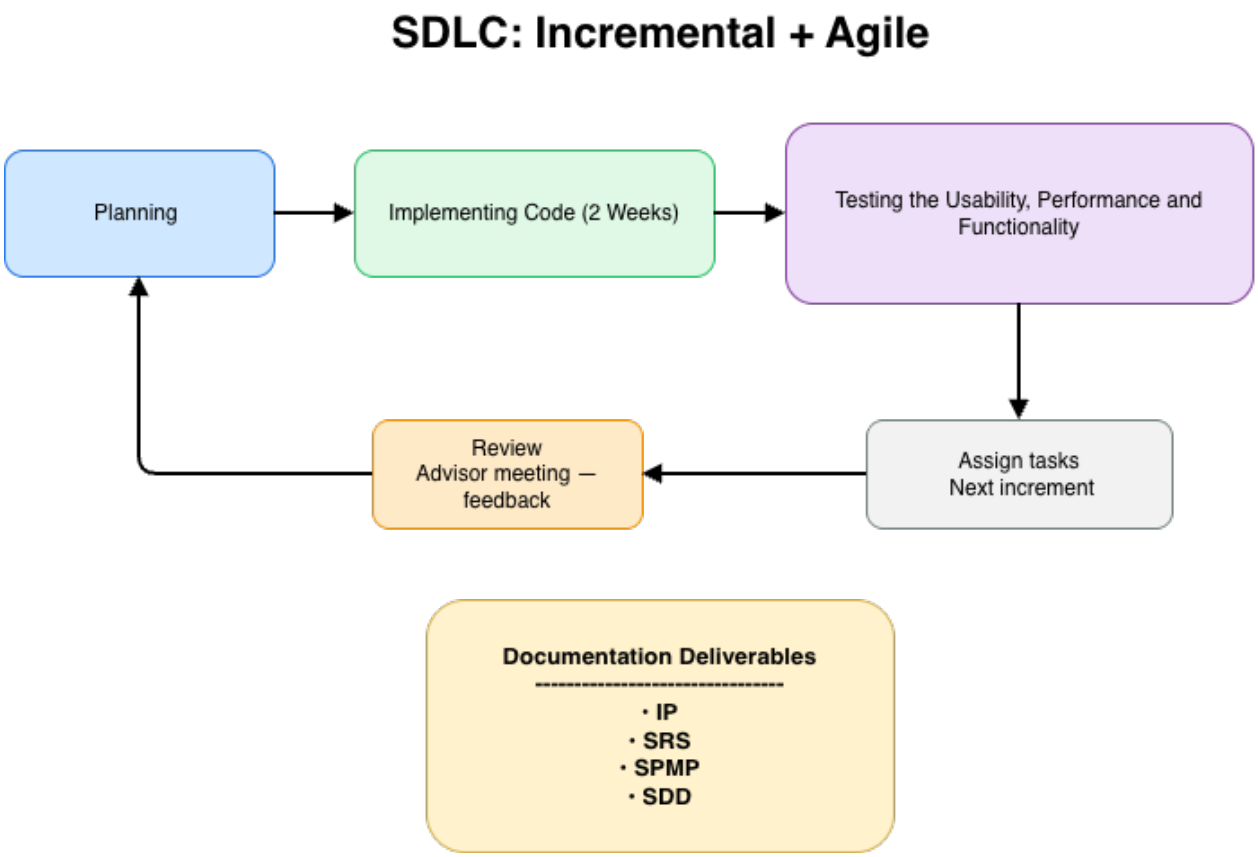

Figure 2 Software Development Life Cycle Diagram

[4] ChatGPT 5 Prompt: **"Write this part again by connecting it with incremental model leading by agile methodology (we sent a previous part of the initial plan)"**

[5] ChatGPT 5-Mini Prompt: **"Write a sentence that summarizes this paragraph."**

4. Project Scope

The ObservAI project encompasses all the technical and academic tasks that will take place from the beginning of CTIS411 up to the end of June 2026. The project shall proceed through the typical steps defined by the CTIS Project Committee. These consist of preparing the Initial Plan (IP), Software Requirements Specification (SRS), Software Project Management Plan (SPMP), and Software Design Description (SDD) writings. [6] These written components shall accompany the software developmental stages First Increment, Second Increment, and Final Increment Products each exemplifying a significant milestone towards the ultimate objective: a data-informed, AI-supported café and restaurant management system.

The project will aim towards establishing an integrated environment from day one where computer vision and AI-driven customer analytics operate together within a single, intuitive web-based platform. The first half of the course will be dedicated to research, requirement analysis, and design of architecture. It would include investigating open-source AI libraries for analysis of a camera, exploring freely available visualization packages, and designing a secure authentication system for managerial users.[7] The second half of the course will be dedicated to implementation, testing, and tuning. All the increments will be constructed one by one, first creating the backend functionality, then adding the AI and dashboard modules, and finally calibrating the interface into a responsive, visually appealing user experience.

Weekly coordination meetings will be held for code reviews, assignment of tasks, and synchronization of documentation for the sake of ensuring progress. Feedback from the supervisor will be ensured continually on the technical as well as design aspects of the project. Each feature will also go through internal testing cycles at every increment completion to ascertain usability, functionality, and performance. Such tests will serve as quality milestones that guarantee weaknesses are found early on and development discipline is maintained throughout both semesters.[8]

The technical scope involves developing a full-stack web application using React for the front-end, Node.js for backend API management, and AI models for computer-vision tasks such as

[6] ChatGPT 5 Prompt: **"Can you add SRS,SDD, SPMP part properly suitable for our school"**

[7] ChatGPT 5 Prompt: "**Give more extra detail about open-source libraries, free camera analysis like starbucks etc**"

[8] ChatGPT 5 Prompt: "**Add test is going to be happened during all school session**"

person detection, demographic estimation, and movement heatmaps. The system focuses solely on camera-based analytics, real-time occupancy monitoring, and AI-generated insights for managerial decision-making.

In terms of resources, the project will employ open-source technologies, public data sets, and free-tier hosting providers exclusively. Collaboration will be facilitated through GitHub, Cursor, and Notion to ensure that code, documentation, and communication are harmonious. Members of the team will utilize established software engineering practices such as versioning, code reviewing, automating tests, and tracing documentation.[9] All features related to AI will be developed using strict privacy terms; there will be no face recognition and personally identifiable information processing. All visible data will be anonymized based on KVKK and GDPR regulations.

Some limitations and constraints define the scope of the project. Time is the most significant constraint with the deliverable to be completed by June 2026. Financial constraints are also there, with the project depending wholly on free or limited open-source-based AI, data storage, and visualization offerings. Live camera integrations, due to limited hardware access, shall be simulated in the form of pre-recorded datasets or public videos. Such simulations, however, will mimic real-world scenarios for testing functional accuracy.

The exclusions in the project strictly identify what is outside the bounds of what is under consideration. ObservAI will not involve mobile applications, facial recognition modules, or direct payment processing. The system will not consist of marketing, deployment, or post-delivery maintenance activities, as the nature of the project is entirely academic and developmental. Similarly, the application will not be taken to other industries apart from the hospitality industry such as retail or logistics. Only a web-based, responsive version of the platform will be developed and turned over.

In short, the ObservAI project scope covers the entire life cycle of a smart, data-centric software solution from idea to proof-of-concept. In the academic timeline, the project will show how new-age cafés and restaurants can leverage artificial intelligence, computer vision, and real analytics to enhance decision-making, optimize efficiency, and reimagine the way small businesses handle day-to-day operations.

[9] ChatGPT 5 Prompt: **"Give some extra good point"**

4.1 Work Breakdown Structure

Breakdown	Description	WBS	Code
ObservAI			1
Deliverable 1	**IP (Initial Plan)**		**1.1**
Activity 1	Conduct problem about café and restaurant management inabilities	1.1.1	
Activity 2	Define project scope, objectives, and limitations	1.1.2	
Deliverable 2	**SRS (Software Requirements Specification Document)**		**1.2**
Activity 1	Collect necessary requirements	1.2.1	
Activity 2	Prepare the finalized requirements specification document	1.2.2	
Activity 3	Conduct advisor review and get feedback	1.2.3	
Deliverable 3	**SPMP (Software Project Management Plan Document)**		**1.3**
Activity 1	Define overall project timeline and milestones	1.3.1	
Activity 2	Identify team roles, tools, and project resources	1.3.2	
Activity 3	Prepare risk management plan and mitigation strategies	1.3.3	

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Deliverable 4	**SDD (Software Design Description Document)**		**1.4**
Activity 1	Design overall system architecture and component interactions	1.4.1	
Activity 2	Design database skeleton (camera analytics, zones, users)	1.4.2	
Activity 3	Create UI frames and dashboard samples	1.4.3	
Deliverable 5	**1st Increment (20%)**		**1.5**
Activity 1	Develop authentication module	1.5.1	
Activity 2	Implement dashboard UI skeleton with placeholder data	1.5.2	
Activity 3	Implement basic zone labeling interface	1.5.3	
Deliverable 6	**2nd Increment (20%)**		**1.6**
Activity 1	Integrate camera AI module (YOLOv8 nano)	1.6.1	
Activity 2	Implement demographic estimation & heatmap generation	1.6.2	
Activity 3	Implement zone-based analytics	1.6.3	
Deliverable 7	**Final Increment (60%)**		**1.7**
Activity 1	Implement AI Insight Engine	1.7.1	

Activity 2	Implement alert & notification system	1.7.2	
Activity 3	Conduct system-wide testing and bug fixing	1.7.3	
Activity 4	Deployment & Demonstration	1.7.4	

4.2 Milestones & Deliverables

Work Products and Delivery Dates for the Project

Below is the list of major deliverables, milestones, and their expected completion dates based on the CTIS411 academic calendar and standard project timeline:

1. **Initial Plan (IP)** — *September 16 – October 6, 2025*
    a. **1.1.1 Conduct problem about café and restaurant management inabilities (Sep 16 – Sep 25):** Identify key operational inefficiencies in cafés/restaurants and evaluate whether camera-based AI analytics can solve them.
    b. **1.1.2 Define the project scope, objectives, and limitations (Sep 26 – Oct 6):** Define ObservAI's boundaries real-time camera analytics, demographic estimation, heatmaps, zone labeling.
2. **Software Requirements Specification (SRS)** — *October 6 – November 10, 2025*
    a. **1.2.1 Collect necessary requirements (Oct 6 – Oct 15):** Gather all functional and non-functional requirements: AI analytics, dashboard, insights, zone labeling, alert system.
    b. **1.2.2 Prepare the finalized requirements specification document (Oct 16 – Oct 30):** Structure FR/NF requirements, use cases, constraints, and database requirements into IEEE SRS format.
    c. **1.2.3 Conduct advisor and get feedback (Oct 31 – Nov 10):** Present SRS to advisor; revise according to feedback for final approval.
3. **Software Project Management Plan (SPMP)** — *November 4 – December 1, 2025*

a. **1.3.1 Define overall project timeline and milestones (Nov 4 – Nov 10):** Create project calendar, define increments, assign milestone deadlines.
b. **1.3.2 Identify team roles, tools, and other project resources (Nov 11 – Nov 20):** Assign responsibilities, determine tools (React, Node.js, YOLOv8 nano), hosting environment.
c. **1.3.3 Prepare risk management plan and mitigation strategies (Nov 21 – Dec 1):** Identify risks (AI accuracy, camera failure, network issues) and define contingency plans.

4. **Software Design Description (SDD)** — *December 2, 2025 – January 5, 2026*
   a. **1.4.1 Design overall system architecture and component interactions (Dec 2 – Dec 15):** Design architecture: AI module (YOLO), backend controllers, dashboard, alert engine.
   b. **1.4.2 Design database skeleton for sales, camera data, and user management (Dec 16 – Dec 23):** Create DB schema for users, cameras, analytics records, zones; ensure real-time analytics compatibility.
   c. **1.4.3 Create UI frames and dashboard samples (Dec 24 – Jan 5):** Prepare UI mockups for authentication, dashboard, demographic charts, heatmaps, zone labeling.
5. **First Increment (20%)** —Estimated: December 5 – *January 8, 2026*
   a. **1.5.1 Develop authentication module (Dec 5– Dec 24):** Implement secure manager login flow (UC-01).
   b. **1.5.2 Implement dashboard UI skeleton with placeholder data (Dec 25 – Jan 8):** Create initial analytics dashboard layout (UC-02).
   c. **1.5.3 Implement basic zone labeling interface (Dec 25 – Jan 8):** Provide simple UI for marking entrance/exit zones (UC-08).
6. **Second Increment (20%)** —Estimated: *Jan 8 – March 13, 2026*
   a. **1.6.1 Integrate camera AI module (YOLOv8 nano) (Jan 08 – Feb 10):** Implement real-time detection, tracking, visitor counting.
   b. **1.6.2 Implement demographic estimation & heatmap generation (Feb 10 – Mar 13):** Add age/gender estimation and movement heatmap visualization.
   c. **1.6.2 Implement zone-based analytics (Feb 10 – Mar 13):** Entrance/exit flow, dwell-time tracking, zone transitions.
7. **Final Increment (60%)** — Estimated: *March 13 – May 25, 2026*

a. **1.7.1 Implement AI Insight Engine (Mar 13 – Apr 15):** Generate behaviour patterns & operational insights (FReq4).
b. **1.7.2 Implement alert & notification system (Apr 15 – May 15):** Real-time crowd/abnormal movement alerts (FReq3).
c. **1.7.3 Conduct system-wide testing and bug fixing (May 15 – May 25):** Full integration tests for AI, backend, dashboard.
d. **1.7.4 Deployment & Demonstration (May 15 – May 25):** Deploy system + prepare final demo materials.

Milestone Summary

- **M1:** Initial Plan Submission – *October 6, 2025*
- **M2:** SRS Submission – *November 10, 2025*
- **M3:** SPMP Submission – *December 2, 2025*
- **M4:** SDD Submission – *January 5, 2026*
- **M5:** 1st Increment (20%) – *January 8, 2026*
- **M6:** 2nd Increment (20%) – *March 15, 2026*
- **M7:** Final Increment (60%) – *May 25, 2026*

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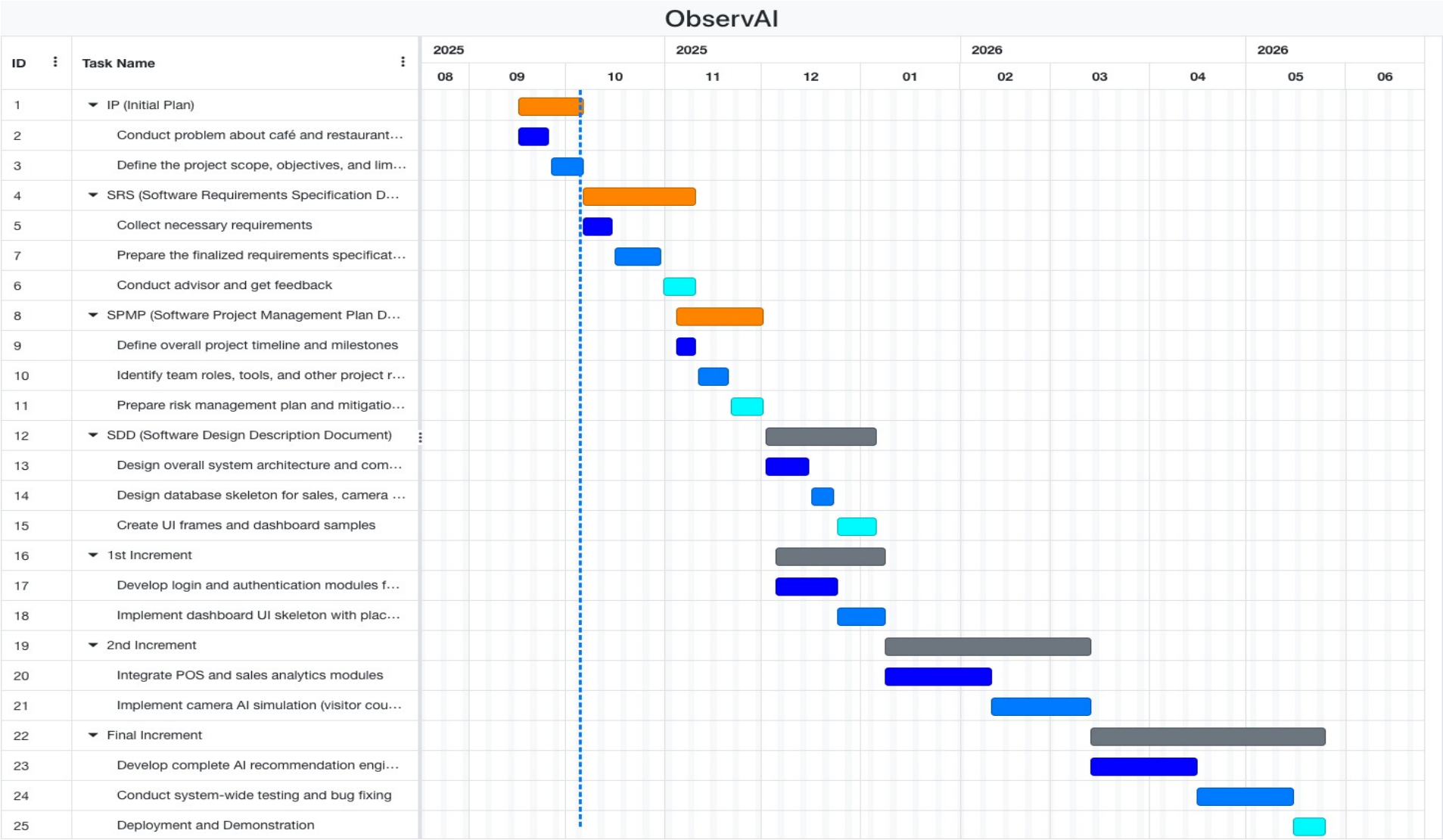

Figure 3 Gantt Chart

5. Project Stakeholders and Organization

Stakeholders of our project consist of team members, advisors, cafés and restaurants. Team members belong to the category of stakeholders due to the fact that the outcome of the project influences our academic results. We should develop the program, implement its functions, and complete tasks in due time. Our advisor assists in offering tips, reviewing documents, and ensuring that the project is aimed at academic objectives. The advisor is more significant in the list of stakeholders since we implement changes in our job in response to their recommendations. Cafés and restaurants remain stakeholders since they will provide feedback and real-world illustrations in support of the test of the system. Cloud hosting services may be considered secondary stakeholders, as they supply the infrastructure that enables the web application to operate during development and testing. There is a straightforward structure of our project: team members collaborate in equal positions, and the advisor offers guidance.

6. Project Communication

Inner-team communication will be carried out through weekly face-to-face meetings, where progress will be discussed and tasks will be assigned. In addition, communication with the project advisor will be maintained via Zoom meetings and WhatsApp messages to ensure guidance and feedback throughout the project. Moreover, WhatsApp will be used for quick daily coordination, while GitHub**,** Notion**,** and Google Docs will serve as platforms for task assignment, documentation, and version control. Meeting notes will be taken regularly to keep track of responsibilities and deadlines. Advisor communication will be maintained through bi-weekly meetings. In these sessions, the team will present updates, receive feedback, and discuss improvements. Notes and deliverables will be shared with the advisor to ensure transparency and alignment with project goals. External communication with cafés, restaurants and cloud providers will be arranged when needed. These interactions will include feedback collection, and integration discussions, with outcomes documented and shared among the team.[10]

[10] ChatGPT 5 Prompt: **"How can we properly characterize our weekly advisor meetings, including how we give updates, request comments, and keep everything on track with project objectives?"**

7. Project Change Control

Change Control for this project will ensure that changes to the code or documents are tracked, reviewed, and accepted in an orderly fashion. The team uses Microsoft Word (Office 365) for collaborative editing and management of project documents. The repository for this project is located on GitHub at https://github.com/observaianeye/ObservAI . Changes to the code will employ GitHub branches and pull requests. Every bug fix or new feature is completed in a separate branch. Before a pull request (PR) is sent to the main branch, it must be reviewed and merged by at least one team member. This timetable keeps the project stable for both semesters, prevents conflicts, and maintains high code quality. Changes to documents will also be documented in the GitHub repository. Any changes to project reports, meeting minutes, or technical documents will be stored with descriptive messages and tied to the code. This provides for version control and tracking changes. Furthermore, GitHub Issues and Trello will be utilized to manage requests for changes and tasks, ensuring that all changes, both non-technical and technical, become accessible to the team. All accepted changes will be recorded and escalated to the correct members to ensure accountability. This consistent method assists the team in handling changes in an orderly fashion while maintaining openness and cooperation throughout the duration of the project.[11]

[11] ChatGPT 5 Prompt: **"Continue this section by describing how the team ensures all changes are visible and well-organized by using Trello and Github Issues."**

8. Assumptions

Technology Availability: It is assumed that open-source AI frameworks and computer-vision libraries (e.g., OpenCV, PyTorch, Ultralytics YOLO) shall remain accessible and stable throughout the development process. Free-tier cloud or local hosting environments will be available for deployment and testing.

Hardware Access: For our case, cafés and restaurants should have cameras installed, thus we will not have extra costs for equipment.

Data Compliance: All visual data collected shall be anonymized immediately after AI processing. The system will conform to KVKK and GDPR principles (data minimization, purpose limitation, right to erasure) without storing any identifiable visuals.

External Cooperation: Restaurants and cafés would, we assume, test prototype implementations of the system and would seek their feedback.[12]

Network Connectivity: A stable local internet connection will be available in testing and demonstration environments to support camera streaming and data synchronization between backend and dashboard.

Advisor Assistance: We anticipate that our advisor will provide frequent assistance and guidance throughout the project.

[12] ChatGPT 5 Prompt: **"Can you give us Assumption part related to both SRS document and Initial Plan First. Can you mention the technology availability part more clearly ?"**

9. Risks

Potential risks we might encounter are:

Risk (R)

R1: Team member unavailability or delay: Tasks may be delayed, so backups and weekly check-ins are used.

R2: Technical difficulties with AI/computer vision libraries: Libraries may fail, so alternatives and early testing are prepared.

R3: Technical Challenges with AI Models – Integrating and optimizing AI-based computer vision models (e.g., crowd detection, heatmaps) may take longer than expected or produce lower-than-expected accuracy.

R4: Cloud service downtime or storage issues: Cloud may fail, so backups and monitoring is used.

R5: Documentation or GitHub mismanagement: Documentation may be lost, so version control and peer review are enforced.

R6: Privacy and Legal Risks – Improper handling of video analytics data may conflict with privacy laws (KVKK/GDPR).

R7: Network or Hardware Issues – Poor camera connections or unstable local networks could interrupt real-time analytics.

How We Identified the Risks

- Review of technical requirements (AI, RTSP, LLM, dashboard, database).
- Examination of dependencies (libraries, cloud, camera hardware).
- Consideration of team structure, workload, and scheduling.
- Analysis of legal constraints (KVKK/GDPR).
- Discussion during meetings with team members and advisor.

9.1 Risk Interpretation

- **Low Risk:** RF < 0.30
- **Medium Risk:** 0.30 ≤ RF ≤ 0.70
- **High Risk:** RF > 0.70

R1: Team Member Unavailability or Delay

Due to exams, illness, workload, or schedule conflicts, team members may fail to complete tasks on time.

Mitigation: Weekly check-ins, backup assignments, workload redistribution.

Early Warning Signals: Missed internal deadlines, reduced communication.

Expected Reaction: Reassign tasks; lower priority items postponed to next increment.[13]

Probability of Failure:

Probability of Failure (Pf)		
Maturity	Complexity	Dependency
0.4	0.5	0.6

Consequence of Failure (Cf)			
Cost	Schedule	Reliability	Performance
0.3	0.5	0.5	0.4

Figure 4 Team Member Unavailability or Delay Risk Calculation

$$Pf = \frac{0.4 + 0.5 + 0.6}{3} = 0.50$$

Consequence of Failure:

$$Cf = \frac{0.3 + 0.5 + 0.5 + 0.4}{4} = 0.425$$

Risk Factor

$$RF = 0.50 + 0.425 - (0.50 \times 0.425) = 0.7125$$

Risk Level: High

R2: Technical Difficulties with AI / CV Libraries

OpenCV, Ultralytics, or PyTorch may fail, be incompatible, or behave unpredictably.

Mitigation: Early prototype, version-lock dependencies, backup alternative models.

Early Warning Signals: Runtime errors, dependency conflicts.

[13] ChatGPT 5 Prompt: **"Can you explain the first risk that I gave in accordance with our SRS and IPv2 documents, explain the risk with adopting those requirements?"**

Expected Reaction: Switch to lighter YOLO models or lower resolution inference.[14]

Probability of Failure:

Probability of Failure (Pf)		
Maturity	Complexity	Dependency
0.7	0.8	0.8

Consequence of Failure (Cf)			
Cost	Schedule	Reliability	Performance
0.3	0.5	0.9	0.9

Figure 5 Technical Difficulties with AI / CV Libraries Risk Calculation

$$Pf = \frac{0.7 + 0.8 + 0.8}{3} = 0.77$$

Consequence of Failure:

$$Cf = \frac{0.3 + 0.5 + 0.9 + 0.9}{4} = 0.65$$

Risk Factor:

$$RF = 0.77 + 0.65 - (0.77 \times 0.65) = 0.919$$

Risk Level: High

R3: AI Model Accuracy Problems

Heatmaps or crowd detection may be inaccurate in noisy environments.

Mitigation: Smoothing filters, additional testing, hyperparameter tuning.

Early Warning Signals: Counting errors > 15%, mismatched heatmaps.

Expected Reaction: Apply post-processing, calibrate cameras.[15]

Probability of Failure:

Probability of Failure (Pf)		
Maturity	Complexity	Dependency
0.7	0.8	0.8

Consequence of Failure (Cf)			
Cost	Schedule	Reliability	Performance
0.5	0.7	0.9	0.7

[14] ChatGPT 5 Prompt: **"Can you explain the second risk that I gave in accordance with our SRS and IPv2 documents, explain the risk with adopting those requirements?"**

[15] ChatGPT 5 Prompt: **"Can you explain the third risk that I gave in accordance with our SRS and IPv2 documents, explain the risk with adopting those requirements?"**

Figure 6 AI Model Accuracy Problems Risk Calculation

$$Pf = \frac{0.7 + 0.8 + 0.8}{3} = 0.77$$

Consequence of Failure:

$$Cf = \frac{0.5 + 0.7 + 0.9 + 0.7}{4} = 0.70$$

Risk Factor:

$$RF = 0.77 + 0.70 - (0.77 \times 0.70) = 0.931$$

Risk Level: High

R4: Cloud Service Downtime or Storage Issues

Cloud delays, outages, or quota limits may stop analytics temporarily.

Mitigation: Local fallback storage, retry logic, monitoring tools.

Early Warning Signals: High latency, partial uploads.

Expected Reaction: Switch temporarily to cached local data.[16]

Probability of Failure:

Probability of Failure (Pf)		
Maturity	Complexity	Dependency
0.3	0.5	0.7

Consequence of Failure (Cf)			
Cost	Schedule	Reliability	Performance
0.5	0.7	0.7	0.5

Figure 7 Cloud Service Downtime or Storage Issues Risk Calculation

$$Pf = \frac{0.3 + 0.5 + 0.7}{3} = 0.50$$

Consequence of Failure:

$$Cf = \frac{0.5 + 0.7 + 0.7 + 0.5}{4} = 0.60$$

Risk Factor:

[16] ChatGPT 5 Prompt: **"Can you explain the fourth risk that I gave in accordance with our SRS and IPv2 documents, explain the risk with adopting those requirements?"**

$$RF = 0.50 + 0.60 - (0.50 \times 0.60) = 0.80$$

Risk Level: High

R5: Documentation or GitHub Mismanagement

Files may be lost, or merges may go wrong.

Mitigation: Mandatory PR reviews, structured repository, weekly backups.

Early Warning Signals: Merge conflicts, unreviewed commits.

Expected Reaction: Rollbacks and stricter Git rules.[17]

Probability of Failure:

Probability of Failure (Pf)		
Maturity	Complexity	Dependency
0.3	0.5	0.3

Consequence of Failure (Cf)			
Cost	Schedule	Reliability	Performance
0.3	0.5	0.3	0.3

Figure 8 Documentation or GitHub Mismanagement Risk Calculation

$$Pf = \frac{0.3 + 0.5 + 0.3}{3} = 0.37$$

Consequence of Failure:

$$Cf = \frac{0.3 + 0.5 + 0.3 + 0.3}{4} = 0.35$$

Risk Factor:

$$RF = 0.37 + 0.35 - (0.37 \times 0.35) = 0.5905$$

Risk Level: Medium

R6: Privacy and Legal Compliance Risks (KVKK/GDPR)

Accidental storage of identifiable images may violate privacy laws.

Mitigation: No raw-frame storage, anonymization at inference layer, privacy checklist.

[17] ChatGPT 5 Prompt: **"Can you explain the fifth risk that I gave in accordance with our SRS and IPv2 documents, explain the risk with adopting those requirements?"**

Early Warning Signals: Image-like data in logs, long buffer times.

Expected Reaction: Immediate refactor & deletion of stored frames.[18]

Probability of Failure:

Probability of Failure (Pf)		
Maturity	Complexity	Dependency
0.5	0.7	0.7

Consequence of Failure (Cf)			
Cost	Schedule	Reliability	Performance
0.9	0.7	0.9	0.7

Figure 9 Privacy and Legal Compliance Risk Calculation

$$Pf = \frac{0.5 + 0.7 + 0.7}{3} = 0.63$$

Consequence of Failure:

$$Cf = \frac{0.9 + 0.7 + 0.9 + 0.7}{4} = 0.80$$

Risk Factor:

$$RF = 0.63 + 0.80 - (0.63 \times 0.80) = 0.926$$

Risk Level: High

R7: Network or Hardware Issues (Camera / RTSP)

Camera may disconnect, local network may be unstable, RTSP streams may drop.

Mitigation: Wired connections, retry & buffering, early camera testing.

Early Warning Signals: FPS drops, "camera offline" logs.

Expected Reaction: Switch to backup feed, keep last frame on screen.[19]

Probability of Failure:

[18] ChatGPT 5 Prompt: **"Can you explain the sixth risk that I gave in accordance with our SRS and IPv2 documents, explain the risk with adopting those requirements?"**

[19] ChatGPT 5 Prompt: **"Can you explain the seventh risk that I gave in accordance with our SRS and IPv2 documents, explain the risk with adopting those requirements?"**

Probability of Failure (Pf)		
Maturity	Complexity	Dependency
0.5	0.5	0.7

Consequence of Failure (Cf)			
Cost	Schedule	Reliability	Performance
0.5	0.7	0.5	0.7

Figure 10 Network or Hardware Issues Risk Calculation

$$Pf = \frac{0.5 + 0.5 + 0.7}{3} = 0.56$$

Consequence of Failure:

$$Cf = \frac{0.5 + 0.7 + 0.5 + 0.7}{4} = 0.60$$

Risk Factor:

$$RF = 0.56 + 0.60 - (0.56 \times 0.60) = 0.824$$

Risk Level: High

9.2 Risk Ranking

Impact ↓ / Probability →	Low (0.0–0.3)	Medium (0.3–0.6)	High (0.6–1.0)
High Impact (0.6–1.0)		R7, R4	R3, R6, R2
Medium Impact (0.4–0.6)		R1	
Low Impact (0.0–0.4)		R5	

Figure 12 Impact Matrix of the Risks

The Impact Matrix was constructed based on the calculated Probability Factor (Pf) and Consequence Factor (Cf) values for each identified risk.

Risks R3 (AI optimization failure), R6 (legal non-compliance), R2 (technical failure of CV libraries) appear in the highest-risk quadrant, indicating both high likelihood and severe impact.

Risks R7 and R4 fall into the medium–high zone, meaning they are reasonably likely and can significantly disrupt the system.

R1 and R5 remain in the medium or medium-low-risk zones, requiring monitoring but not representing immediate threats.[20]

[20] ChatGPT 5 Prompt: **"Explain this figure according to our project risks and their impact on the project"**

10. Discussions

We experienced a few problems when working on the first strategy for ObservAI, most of which included arranging meetings because everyone had busy weeks. Despite this, we spoke often, primarily via WhatsApp. We are planning to use Notion and Trello for sharing information, assigning work, and maintaining an even schedule. [21]There was never any health, safety, law, or moral issue in the process of the project, as we proceeded in a responsible and respectful approach and only followed rules of privacy. Until now, we in our group have taken care of the whole project and never had to employ outside help. Overall, these experiences showed that flexible planning, regular communication, and effective organization of tasks is valuable in ensuring the smooth flow of the project.

[21] ChatGPT 5 Prompt: **"Can you rewrite my paragraph about how our team handled early communication issues and how we plan to stay organized moving forward."**

11. Curriculum Vitae

FAIK EMRE PARTAL
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EDUCATION

Bilkent University — **ANKARA, TURKEY**
Bachelor of Science in Information Systems and Technologies (CTIS) — *2022-PRESENT*

- Transferred from Bayreuth University, Germany (Informatik) after two years of study.
- Expected Graduation: Spring 2026

WORK EXPERIENCE

JOTFORM
Started as a summer intern in the BrandKit project and was transitioned to a term internship as part of the Ignite Jr. team, focusing on backend development and AI-driven features.

Backend Developer Intern (Ignite Jr. Team) — *Sep 2025-Present*

- Building AI-powered suggestion systems for Jotform Inbox using **RAG (Retrieval-Augmented Generation)** with **Qdrant vector database**.
- Designed **PHP** endpoints and integrated embedding pipelines to enhance user comment/reply recommendations.
- Contributing to AI system evaluation through **Postman-based test automation and performance monitoring** of backend endpoints.

Backend Developer Summer Intern (Phoenix Jr. Team) — *Aug 2025 – Sep 2025*

- Developed backend modules for the **BrandKit** project, enabling automatic logo, color, and font extraction from user-entered URL's.
- Implemented **API integrations** and optimized **PHP-based data processing** for faster brand recognition workflows.
- Collaborated with the frontend intern to support **AI-powered brand suggestion** features.

SCALR PLATFORM — *(Remote)*, **HQ: BEAVERTON, OR**
Product Analyst Intern — *June 2022-Sep 2024*
*Scalr is a data-driven operations platform serving coffee and QSR chains such as **Blank Street Coffee**, **Clutch Coffee Bar**, and **Watchhouse**.*

- Supported early product development by analyzing tools for **ordering, inventory, payroll, and finance**.
- **Automated** expense tracking via Ottimate / PlateIQ APIs, identifying abnormal credit-card transactions.
- Drafted **technical requirements** and helped shape features like payroll workflows and finance **integrations**.
- Collaborated with operations teams to surface cost insights and improve financial visibility.
- Standardized internal tools into the **Scalr Cockpit** platform for multi-client deployment.

PROJECT PORTFOLIO

- **Comparative Analysis of English & Turkish Football Leagues**
  - Conducted comparative performance analysis between English Premier League and Turkish Super Lig.
  - Collected + cleaned multi-season data, calculated xG-based metrics, and visualized tactical trends in R Studio.
- **Football Team Simulation in Java**
  - Built a football management simulation with **player rating** and **formation-based strength** mechanics. Planned project goals, define data inputs, and design simulation logic.
  - Implemented realistic match logic and collaborated with a 4-person team for game-balance testing.

TECHNICAL SKILLS & CERTIFICATIONS

- **IELTS 6.5** – English Proficiency
- **TestDaF B2-C1** – German Proficiency
- **Programming Languages:** Java, Python, SQL, C, R, Kotlin
- **Web Development:** HTML, CSS, JavaScript, Node.js, React, PHP
- **Database Management:** Oracle SQL Developer
- **Data Analysis & Visualization: R** Studio, Excel, Tableau, Power BI
- **Business & Product Tools:** JIRA, Confluence, Trello, Figma, Notion, Asana

EXTRACURRICULAR ACTIVITIES

- **Professional Basketball Referee** – *Turkiye Basketbol Federasyonu (TBF)*
- **Basketball Player** – Licensed, semi-professional -- *Turkiye Basketbol Federasyonu (TBF)*
- **Peak Performance Society** (Bilkent University)– *Co-founder/Vice President & Events Director*

HALIL İBRAHIM ATALAY

CV

Ankara +905531851086 atalayhalil155@gmail.com

With over three years of experience as an active and dedicated CTIS student at Bilkent University, I have continuously developed my ability to think critically and solve problems effectively. I am passionate about creating innovative solutions, especially in web development environments. My understanding of programming principles and my ability to work in collaborative teams allow me to adapt to new challenges and exceed expectations. I am committed to personal and professional growth, always striving to deliver high-quality work and contribute to dynamic, forward-thinking environments

Education and Qualifications

CTIS (Computer Technology and Information Systems) Sept 2021 - Jun 2026
Bilkent University, Ankara
-3.19 GPA
-Honor Student Actively

Work experience

Software Engineer Intern Aug 2024 - Sept 2024
Mobiliz Bilgi ve İletişim Teknolojileri A.Ş, Ankara
During my internship at Mobiliz, I had the opportunity to learn backend development in a hands-on, real-world environment. My focus was on understanding and applying core technologies such as Spring Boot, PostgreSQL, and RESTful APIs under the guidance of experienced developers.During my internship at Mobiliz, I had the opportunity to learn backend development in a hands-on, real-world environment. My focus was on understanding and applying core technologies such as Spring Boot, PostgreSQL, and RESTful APIs under the guidance of experienced developers.

Software Engineer Intern Jan 2025 - May 2025
İnnova Bilişim Çözümleri, Ankara
During my 4-month internship at Innova R&D, I worked as a full-stack development intern in the TTVPN project team. My responsibilities included backend and frontend development, bug fixing, documentation, and self-guided research on emerging technologies.

-Developed 11+ live pages using Spring Boot, React, PostgreSQL, DBeaver, and Postman
-Fixed frontend/backend bugs based on QA reports
-Researched and studied Docker and Microservice architecture in parallel
-Worked with legacy codebases (PrimeFaces, JSF) to refactor and modernize UI logic

This internship helped me strengthen my skills in full-stack development, debugging, and agile teamwork in an enterprise-grade environment.

Skills

Java-Spring Boot	Experienced
Microservices	Experienced
Primeface-JSF	Experienced
HTML-Css-Tailwind	Experienced
MySQL-Postgre SQL-Oracle SQL	Experienced
React.js	Experienced
C/C++	Experienced
OpenGL-GLUT	Experienced

Languages

ENGLISH	B2

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Mevlüt Yılmaz

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Education

Bachelor of Information Systems and Technologies — **2022 - 2026 (expected)**
Bilkent University

English Preparatory Program — **09/2022 - 01/2023**
Bilkent University

Experiences

Robit Teknoloji - Intern — **07/2023 - 08/2023**
Software testing for Internet of Things (IoT) with Python, Flutter.

Creasaur Entertainment Co. - Game Developer Intern — **06/2025 - Present**
Developed a solver tool for a puzzle game to assess the level difficulty and playability.

Certifications

Blockchain 101 — **09/2023**
Blockchain101 Camp Completion Certificate.

The Complete Node.js Developer Course (Udemy) — **01/2025**
Certificate of Completion. Learned back-end development using Node.js, including REST APIs and database integration.

Skills

Programming	Java, JavaScript, Node.js, PHP, HTML, CSS, C, C#
Communication	Turkish (native), English (business)
Other	Linux, GitHub, Microsoft Office, Unity Game Development

Initial Plan

Team 12

Semih Şengül
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GitHub Profile
LinkedIn Profile

Education

- **Bilkent University (English)** — *Computer Technology and Information Systems*
  Fourth Year Student — CGPA/Percentage: 3.00

Experience

Creasaur Game Developer — *3 June - Still Working 2025*
Unity 2D and 3D Game Development Intenship

- **Summer Internship in Layermark** — *July-August 2024*
  Frontend Dev "Vue.js"
- **Bilkent Game Development and Animation Club Gamejam** — *16-18 February 2024*
  2D Game Dev
- **Blockchain101 Participation Certificate** — *25 August - 3 September 2023*
  Mustafa Akgül Özgür Yazılım Kampı

Technical Skills and Interests

Languages: Programming in Unity, C/C++, Java, Javascript, Python, R (Completed term projects for each programming language in University's Lecture)
Coursework: Android Studio (Kotlin), PL/SQL ,Cisco Network Course Certificate, Linux, OpenGL
Areas of Interest: Artificial Intelligence, Data Science, Game Development, Guitar Playing, German Language, Bilkent Swimming Club Member , Mathematics (taking additional mathematics lectures)

Initial Plan

Team 12

UĞUR AYBERK KARAKIŞ

SENIOR CTIS STUDENT AT BILKENT UNIVERSITY

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Çankaya , Ankara, Turkey

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SKILLS

- C
- Python
- Tensorflow
- Linux
- HTML
- CSS
- JavaScript
- PHP
- Java
- Kotlin
- SQL
- R
- OpenGL
- GLUT
- Office Tools

LANGUAGES

- English (Fluent)
- French (Beginner)
- German (Beginner)

PROFILE

I am a senior undergraduate student at Bilkent University, studying Computer Technologies and Information Systems. I am passionate about developing innovative software solutions, particularly in the areas of object oriented, back end, database management and administration. In addition, I have experience in media, having hosted programs on Radyo Bilkent.

EDUCATION

High School — 2018 - 2022
Betül Can Anatolian High School

Undergraduate Degree — 2022 - PRESENT
Faculty of Applied Sciences | İhsan Doğramacı Bilkent University
GPA: 3.30 / 4.0

WORK EXPERIENCE

Huawei — 2025 September - Present
IAD Intern
- Currently working about CTR Prediction models.

Radyo Bilkent — 2024 - Present
News Anchor
- Currently presenting live news bulletins and hosting a weekly arts and culture radio program.

MİA Teknoloji — 2025 June - 2025 July
AI Intern
- Coding, developing artificial intelligence models, image processing data cleaning.

Radyo Bilkent — 2023 September-2024January
News Department Intern
- Wrote hourly news bulletins and created content for art and culture programs